

Table VS5. Paired comparison of the molecular representations on the held-out test set.

Each row compares two representations with the classifier held fixed, using the predictions of both models for the same 201 held-out compounds. *Discordant* gives the number of compounds that only the fine-tuned GROVER model classified correctly, out of all compounds the two models classified differently. The p values come from a two-sided exact McNemar test, and the final column applies the Holm step-down correction across all six tests. This table is the one reproduced as Table VS5 in the response to the referees.

Reference representation	Classifier	Fine-tuned GROVER accuracy (%)	Reference accuracy (%)	Discordant	Exact p	Holm-ad p
Pretrained GROVER	Random Forest	92.54	86.07	17 of 21	0.0072	0.028
Pretrained GROVER	XGBoost	93.53	85.57	20 of 24	0.0015	0.007
Pretrained GROVER	MLP	91.54	82.59	21 of 24	0.0003	0.001
ECFP4	Random Forest	92.54	90.55	8 of 12	0.3877	0.775
ECFP4	XGBoost	93.53	89.55	9 of 10	0.0215	0.064
ECFP4	MLP	91.54	89.55	9 of 14	0.4240	0.775

Fine-tuned GROVER was more accurate than pretrained GROVER with every classifier, and all three differences remained below 0.05 after correction. Against ECFP4 the matched differences of 2.0 to 4.0 percentage points rested on 10 to 14 discordant compounds, and none remained below 0.05 after correction, so these tests establish neither an advantage nor equivalence.